Mental-health emergency visits, community follow-up and acute psychiatric beds across Chile’s Health Services, 2024

C1 · Business Intelligence (IIB423T-1) · Universidad del Desarrollo
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1. Summary

In 2024 Chile’s public emergency network recorded 305,431 mental-health visits (1.57% of 19.4 million emergency visits) and 9,753 self-harm visits. The same year, 1,130 acute psychiatric beds were available on average in public hospitals, at 89.3% occupancy, and primary and specialty teams recorded 5.3 million mental-health follow-up visits.

This report compares the 29 Health Services on these three parts of the pathway: emergency visits, community follow-up and acute psychiatric beds. The purpose is to help a Health Service decide which part to review first. The comparison is descriptive and uses one year.

Main findings:

- The two highest mental-health shares of emergency visits (Metropolitano Norte 3.6%, Aconcagua 3.0%) come from specialised psychiatric emergency departments that receive referrals. Without them both Services are in the typical range (1.7% and 1.2%).
- Community follow-up per mental-health emergency visit ranges from 8.4 to 32.6 (median 18.8). Acute occupancy ranges from 41.5% to 99.6%.
- Five Services are above the median on emergency share, below it on community follow-up and at or above 90% occupancy: Biobío, Araucanía Sur, Los Ríos, Osorno and Del Reloncaví. Three Services have none of the three flags.
- Waiting for a bed cannot be measured: the SADU waiting fields are recorded as zero for every facility-day in 2024.

The W1 “stress ratio” is kept only to reconcile W1 figures; it divides visits by bed-days and cannot show a shortage.

2. Problem, foundation and evolution from W1

2.1 Problem and scope

Element	Definition in C1
Problem	Mental-health crises reach emergency departments; whether a Service should first review community follow-up, acute bed capacity or how emergency and hospital care are organised is not visible from any single source
Population	Emergency visits with SADU mental-health causes (F00–F99 and R45.8, cause 36) and self-harm (X60–X84, cause 35); all age bands, reported with SADU labels
Place	29 public Health Services (*Servicios de Salud*); public network only
Period	Calendar year 2024 (closed year)
Units	Visits, hospitalisations, bed-days and follow-up visits; **not people**
Proposed use	A descriptive profile and screen that tells a Health Service which area to review first; not a performance ranking

What “ED” means in this report. ED stands for emergency department: a place where people receive urgent care without an appointment. SADU covers the whole public emergency network, so “ED visits” means

visits to any of its facilities:

- hospital emergency units (*Unidad de Emergencia Hospitalaria*, UEH), 182 facilities, the only ones with direct access to hospital beds;
- primary-care urgent centres without beds: SAPU (*Servicio de Atención Primaria de Urgencia*), SAR (*Servicio de Alta Resolutividad*) and SUR (*Servicio de Urgencia Rural*), plus 3 CEAR, 458 facilities in total.

“Hospital ED” refers only to the first group. A psychiatric ED is a hospital emergency unit specialised in mental health, such as those of Instituto Psiquiátrico Dr. José Horwitz Barak and Hospital Psiquiátrico Dr. Philippe Pinel.

2.2 Documentary support

Each source is used for the claim it supports. Policy documents give context; they are not used as evidence that a Service fails a legal duty.

- **Policy objective.** The National Health Strategy toward 2030 (MINSAL, 2022 edition) sets the impact objective “Mejorar la salud mental de la población en Chile con enfoque en lo promocional y preventivo” (p. 35). Its mental-health section states expected results (e.g. more coverage of promotion and prevention actions with an equity focus) but **no numeric target**. The W1 report stated a “15% reduction in youth suicide” target; it is not in the document and has been removed.
- **Legal context.** Law 21.331 (Diario Oficial, 11 May 2021) defines psychiatric hospitalisation as an exceptional and transitory resource, used when outpatient treatment is not possible, and prohibits new asylum-type psychiatric institutions. It motivates looking at community follow-up next to beds. It is not used here to judge compliance.
- **Financing context.** The University of Chile School of Public Health report for MINSAL (2024) estimates mental-health spending at 4.89% of total health spending in 2022, excluding sick-leave subsidies. The W1 figure of “about 2.4% of the budget” had no source and has been removed.
- **Emergency-to-hospital pathway.** González Valderrama et al. (2023) studied 90,464 psychiatric emergency visits at Instituto Psiquiátrico Dr. José Horwitz Barak (2017–2020): 12.5% led to an indication for hospitalisation, and 35.9% of those indications could not be carried out for lack of beds. The study separates visits, decisions to hospitalise and admissions; C1 follows the same separation with public data.

Level	Example	Used in C1 as
Policy objective	ENS 2030 mental-health objective	Context for the decision
Numerical target	None in the ENS mental-health section	Not applicable
Indicator our data measure	Mental-health share of emergency visits, follow-up visits per emergency visit, acute occupancy	Descriptive comparison between Services

2.3 Alternatives in W1 and why this line was kept

W1 compared two lines: (1) seasonal respiratory visits and critical-care bed occupancy, and (2) mental-health emergency visits and psychiatric beds. Line 2 was preferred because it links a documented pathway problem (emergency visits that need a psychiatric bed, per the Horwitz study) with three public sources that share a territorial code. Line 1 remains feasible, but most of its value depends on comparable years to study seasonality, and it was already a common topic in the course.

The instructor’s feedback gave line 2 a clearer purpose: compare community care, emergency visits and hospital use to show which part of a network deserves closer attention. C1 adopts that purpose. Section 8.2 records each piece of advice and the response.

3. Seven-step map

Step	C1 content	Where
1. Decision	**Proposed user:** the mental-health unit of a Health Service, or the MINSAL network division that compares Services. **Decision:** which area to review first. **Alternatives:** (a) community follow-up capacity, (b) acute psychiatric bed capacity and management, (c) the organisation of emergency and referral care, (d) no action	§2.1
2. Success criteria	**Desired outcome:** crises treated in time and at the right level, admissions without long waits. **Measurable now:** each Service's position against the other 28 on five indicators. **Not measurable now:** waiting time, clinical outcomes, number of people	§3, §6
3. Business questions	**Main:** in 2024, how did the 29 Services differ in mental-health emergency visits, community follow-up and acute psychiatric bed use, and which area deserves a closer look in each? **Supporting:** Q1 composition by cause and age; Q2 concentration in specialised EDs; Q3 variation in follow-up and bed use; Q4 which Services combine flags; Q5 how sensitive the result is to definitions	§5
4. Business process	Crisis → primary care or COSAM → emergency visit (SAPU, SAR, SUR or hospital ED) → decision to hospitalise → wait for a bed → admission to an acute psychiatric area → discharge → community follow-up. Observed: visits, ED hospitalisations, bed-days, discharges, follow-up visits. **Not observed:** waiting (recorded as zero), transfers between Services, the private sector, patients' residence, outcomes	Fig. A
5. Entities, events, states	Entities: facility, Health Service. Events (aggregated counts): emergency visit, ED hospitalisation, discharge, follow-up visit. State: bed-day available or occupied. No person identifier in any source	§4.1
6. Data representation	See §4.1: one row per facility-day-cause (SADU), hospital-area-month (REM20) and facility-month-activity (REM A06). Joins: SADU facility code → registry (N:1); all three sources → Health Service code (aggregation, not row joins)	§4.3
7. Measurement	Indicator definitions with numerator, denominator and unit in §4.5	§4.5

Figure A. Pathway stages and what each source observes. No source carries a patient identifier, so the stages cannot be linked record by record.

Stage	Source and field	Observed in 2024
Community care	REM A06 A.1 follow-up visits	Yes: visits per facility-month

Stage	Source and field	Observed in 2024
Emergency visit	SADU causes 36 and 35	Yes: visits per facility-day
Decision to hospitalise from the ED	SADU cause 42	Yes: hospital EDs only, any ward
Waiting for a bed	SADU causes 27 and 28	No: recorded as zero everywhere
Acute psychiatric stay	REM20 bed-days and discharges	Yes: per hospital-area-month
Follow-up after discharge	REM A06 A.1	Only as total volume, not linked to discharges

4. Data, preparation and quality

4.1 Sources and observation units

Source (version)	One row =	Rows used	Key fields
SADU, Atenciones de Urgencia 2024 (DEIS download)	facility × day × cause, counts by 5 age bands	4,037,964 extracted from 8,973,229	`IdEstablecimiento`, `IdCausa`, `fecha`, `CodigoDependencia`
REM20, `indicadores_rem20_20260825.csv`	public hospital × functional area × month	745 acute rows of 934 psychiatric rows (2024)	`CODIGO_ESTABLECIMIENTO`, `COD_AREA_FUNCIONAL`, `MES`, `COD_SSS`
REM Serie A 2024, sheet A06 section A.1	facility × month × professional code	72,522 of 6,559,080	`IdEstablecimiento`, `CodigoPrestacion`, `Mes`, `IdServicio`
DEIS facility registry, snapshot 2026-09-10	facility	5,743	`EstablecimientoCodigo`, `EstablecimientoCodigoAntiguo`

`analysis/extract_c1_inputs.py` filters the original files on one field and copies each line unchanged. Every correction happens in the notebook and is counted there.

4.2 Profiling results that changed the analysis

Cause hierarchy (SADU). SADU mixes totals and components. On all 224,330 facility-days, cause 36 equals the sum of causes 37–41, so 36 is the mental-health visit total and **includes suicidal ideation (R45.8)**. Self-harm (35) is an injury code outside 36 and is kept separate. Cause 42 is not a visit cause: together with 7, 8, 22, 23, 32 and 33 it sums exactly to cause 25 (hospitalisations decided in the ED) on every facility-day. It is the number of ED hospitalisations for mental disorders.

W1 added causes 35, 37, 39 and 40 and called them “acute” visits. That mixed a component of 36 with an external-cause code and left out substance-use and other disorders. C1 uses 36, plus 35 separately.

Recorded zeros versus missing records. Causes 27 and 28 (“patients waiting for hospitalisation”) are zero on every facility-day. These are recorded zeros, not observed waiting, so waiting is outside the analysis. 452 of 640 facilities report all 366 days; 40 report fewer than 300. A missing day is left missing; SADU does not say whether it means a closed service or a reporting gap, and imputing visits would create unobserved counts.

Acute psychiatric areas (REM20). Medium- and long-stay and forensic areas have mean stays of 34–1,379 days and follow other referral routes. C1 uses the four acute areas: adult and child/adolescent short stay, and adult and child/adolescent psychiatric intensive care (69.8% of psychiatric bed-days). W1 used adult long stay and omitted psychiatric intensive care.

4.3 Preparation and joins

Step	Rows or units affected	Effect
Strip spaces in `IdEstablecimiento`	28 facility codes	Needed for the registry join
Drop fully identical duplicate rows	24 rows (2 facility-days)	Mental-health visits 305,432 → 305,431
Fill blank territorial fields of `17-803` (SAPU Ultraestación) from the registry	3,474 rows, 1 facility	Assigned to Ñuble (code 17)
Keep acute psychiatric areas	745 of 934 REM20 rows	Defines the bed measure
Map REM A06 facilities to registry care level	2,296 facilities, 100% found	Splits follow-up into primary care and specialty/other

Facility join: 640 versus 609. W1 reported 640/640 matches in the report and 609 in the notebook. Both came from the same data. In 2024, 609 SADU facilities report the legacy code (`NN-NNN`, after stripping spaces) and 31 report the modern 6-digit code. Looking up both code systems matches all 640, each to exactly one registry facility (expected cardinality N:1), and no registry facility is reached by two SADU codes. The join therefore neither drops nor duplicates rows. The registry contains 13 legacy codes converted to dates by a spreadsheet (e.g. `12-Jan`); none is used by SADU.

Territorial labels. SADU, REM20 and REM A06 use the same 29 numeric Health Service codes but different names (SADU “Iquique” = REM20 “Tarapacá”; “Valdivia” = “Los Ríos”; four more differ in wording). C1 aggregates each source by code and uses REM20 names for display (`data/health_service_labels.csv`). Two facilities (SAPU Padre Esteban Gumucio, SUR Navidad) belong to a different Service in SADU 2024 than in the 2026 registry. C1 keeps the 2024 SADU assignment, since that is where the visits were reported; these facilities hold 0.12% of mental-health visits. Checking network membership against the DEIS 2024 establishment list could not be done from our environment (section 7.3).

Sources are compared, not row-joined. SADU has no bed field and REM20 has no diagnosis, and 458 of 640 emergency facilities (72%: SAPU, SAR, SUR, CEAR) are not hospitals. The three sources are aggregated to Health Service × 2024 and shown side by side.

4.4 Five Cs

C	Check	Result and decision
Clean	Nulls, negatives, age cross-sum, duplicate keys, key whitespace; REM20 duplicates; A06 `Col01` vs sex split	No negatives; age bands sum to `Total` on every row; 24 duplicate SADU rows removed; 28 codes stripped. REM20: 7 zero-bed months, 14 rows with occupied > available kept as reported. A06: 5 null rows left missing; 7 rows (177 visits) where `Col01` ≠ `Col02+Col03` kept with the published total
Consistent	Cause hierarchy on every facility-day; occupancy recomputed from bed-days; months reported per hospital area	Hierarchy holds on all facility-days. Recomputed occupancy matches `INDICE_OCUPACIONAL` within 0.1 pt on all rows. 57 of 64 hospital-area units report 12 months. One year only, so stability across years is not assessed
Conformed	Facility codes across two code systems; Health Service codes and names across three sources	640/640 facilities matched N:1; identical code sets across sources, names differ and are mapped by code
Current	Source dates against the decision window	2024 is the latest closed year in all three sources. The registry is a 2026 snapshot, used only to bridge codes and care level. Suitable for a planning review, not for real-time bed management
Comprehensive	Coverage of facilities, days, sectors and fields	Public network only (REM20 `TIPO_PERTENENCIA` = 1). Waiting is not observed. People, residence, private clinics and transfers between Services are not observed. 188 facilities miss at least one day

4.5 Indicators

Indicator	Numerator	Denominator	Unit	Source
Mental-health share of ED visits	SADU 36 visits	SADU 1 visits (all causes)	% of visits	SADU
Same, excluding psychiatric-hospital EDs	SADU 36 without 2 facilities	SADU 1 without the same	% of visits	SADU

Indicator	Numerator	Denominator	Unit	Source
ED hospitalisations for mental disorders	SADU 42	SADU 36 visits at hospital EDs	per 100 visits	SADU
Average acute psychiatric beds	Available acute bed-days	366	beds	REM20
Acute psychiatric occupancy	Σ occupied acute bed-days	Σ available acute bed-days	%	REM20
Community follow-up intensity	REM A06 A.1 follow-up visits	SADU 36 visits	visits per ED visit	A06, SADU

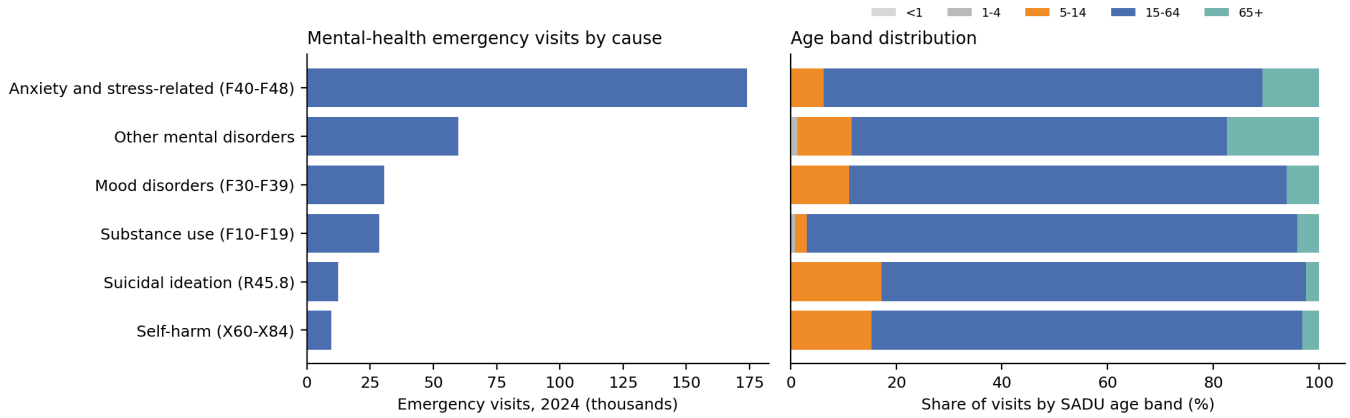
No population rates are used: visits are counted where care took place, which is not the patient's residence, and no population file with the same territorial unit is part of the case. The follow-up indicator combines two sources; it describes activity volume, not the share of emergency patients who were followed up.

5. Results

5.1 National picture

Measure (2024)	Value	Unit
All emergency visits (SADU 1)	19,436,048	visits
Mental-health visits, F00–F99 + R45.8 (SADU 36)	305,431	visits
— at hospital emergency departments	158,027	visits
Self-harm visits, X60–X84 (SADU 35)	9,753	visits
ED hospitalisations for mental disorders (SADU 42)	27,306	hospitalisations
Discharges from acute psychiatric areas (REM20)	16,427	discharges
Available acute psychiatric bed-days (REM20)	413,486	bed-days
Mental-health follow-up visits (REM A06 A.1)	5,315,053	visits

Mental-health visits are 1.57% of emergency visits. For every 100 mental-health visits at hospital EDs there were 17.3 ED hospitalisations for mental disorders. This is not directly comparable with the Horwitz study's 12.5%: that study counts indications at one psychiatric ED, while SADU 42 counts hospitalisations at all hospital EDs, to any ward. There were more ED hospitalisations for mental disorders (27,306) than discharges from acute psychiatric areas (16,427). Possible reasons include admissions to general wards, transfers between hospitals and hospitalisations registered in both sources under different rules. The data cannot tell these apart. Primary care delivers 75% of follow-up visits (4.0 million) and specialty and other levels 25% (1.3 million).



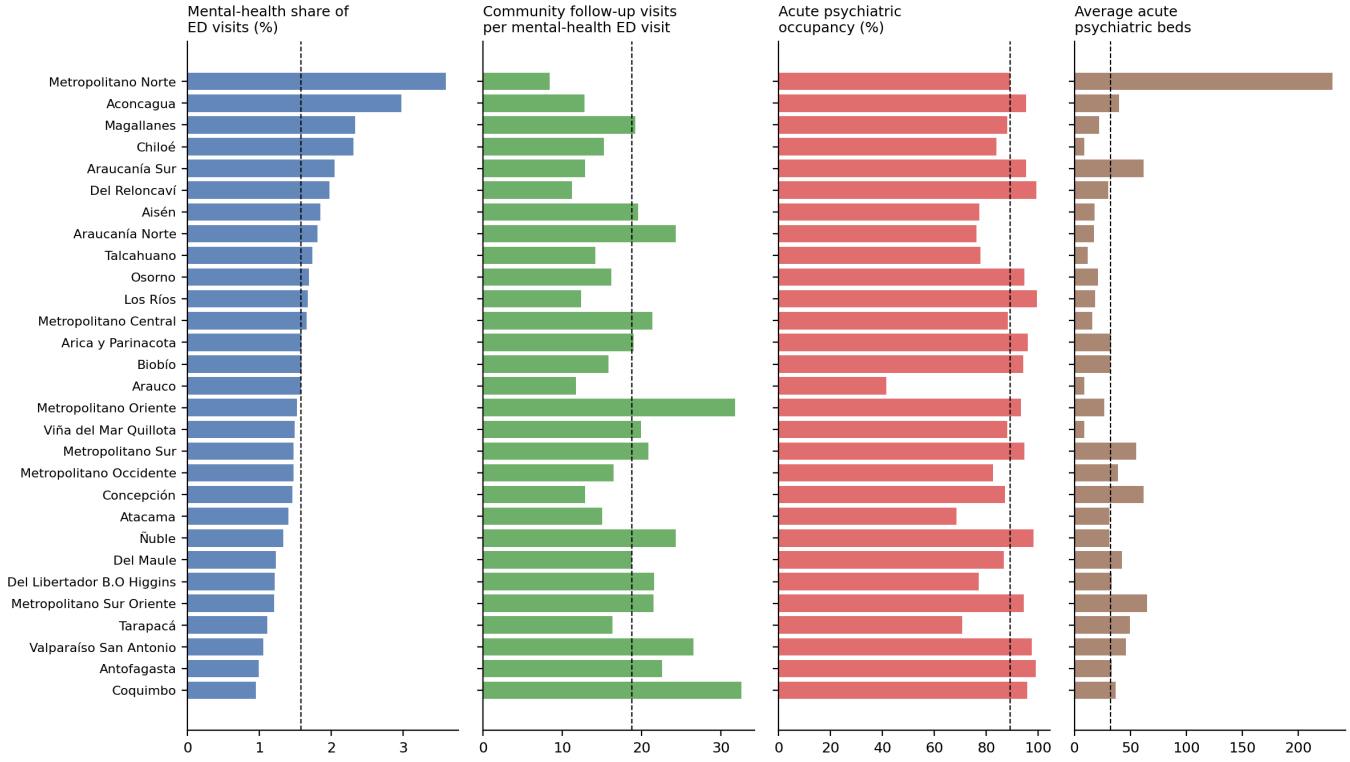
Source: DEIS Atenciones de Urgencia 2024. Visits, not people. 15-64 is the narrowest adult band SADU publishes.

Figure 1. Mental-health emergency visits by cause and SADU age band, 2024.

Anxiety and stress-related disorders (F40–F48) account for 57% of mental-health visits. Suicidal ideation and self-harm together are 7.0% of the combined 36 + 35 total. The 15–64 band holds most visits in every cause. The 5–14 band's share is larger for suicidal ideation (17.2%) and self-harm (15.3%) than for any other cause (2.3–10.9%). The 15–64 band includes adolescents aged 15–17 and cannot be split with these totals, so no adult-only figure is reported.

Mental-health visits per day ranged from 740 (May) to 892 (November), and acute occupancy from 85.3% (December) to 91.4% (October). With one year there is no basis for a seasonal claim (Figure 4, annex).

5.2 Health Service profiles



Dashed line: median of the 29 Health Services. Sources: DEIS SADU 2024, REM20 2024 (acute psychiatric areas), REM A06 section A.1 2024.

Figure 2. Four indicators by Health Service, 2024. Dashed line: median of the 29 Services.

Indicator	Min	Median	Max
Mental-health share of ED visits (%)	0.95 (Coquimbo)	1.58	3.59 (Metropolitano Norte)
Follow-up visits per mental-health ED visit	8.4 (Metropolitano Norte)	18.8	32.6 (Coquimbo)
Acute psychiatric occupancy (%)	41.5 (Arauco)	89.2	99.6 (Los Ríos)

Indicator	Min	Median	Max
ED hospitalisations per 100 MH visits at hospital EDs	0.7 (Araucanía Norte)	21.1	43.9 (Metropolitano Oriente)

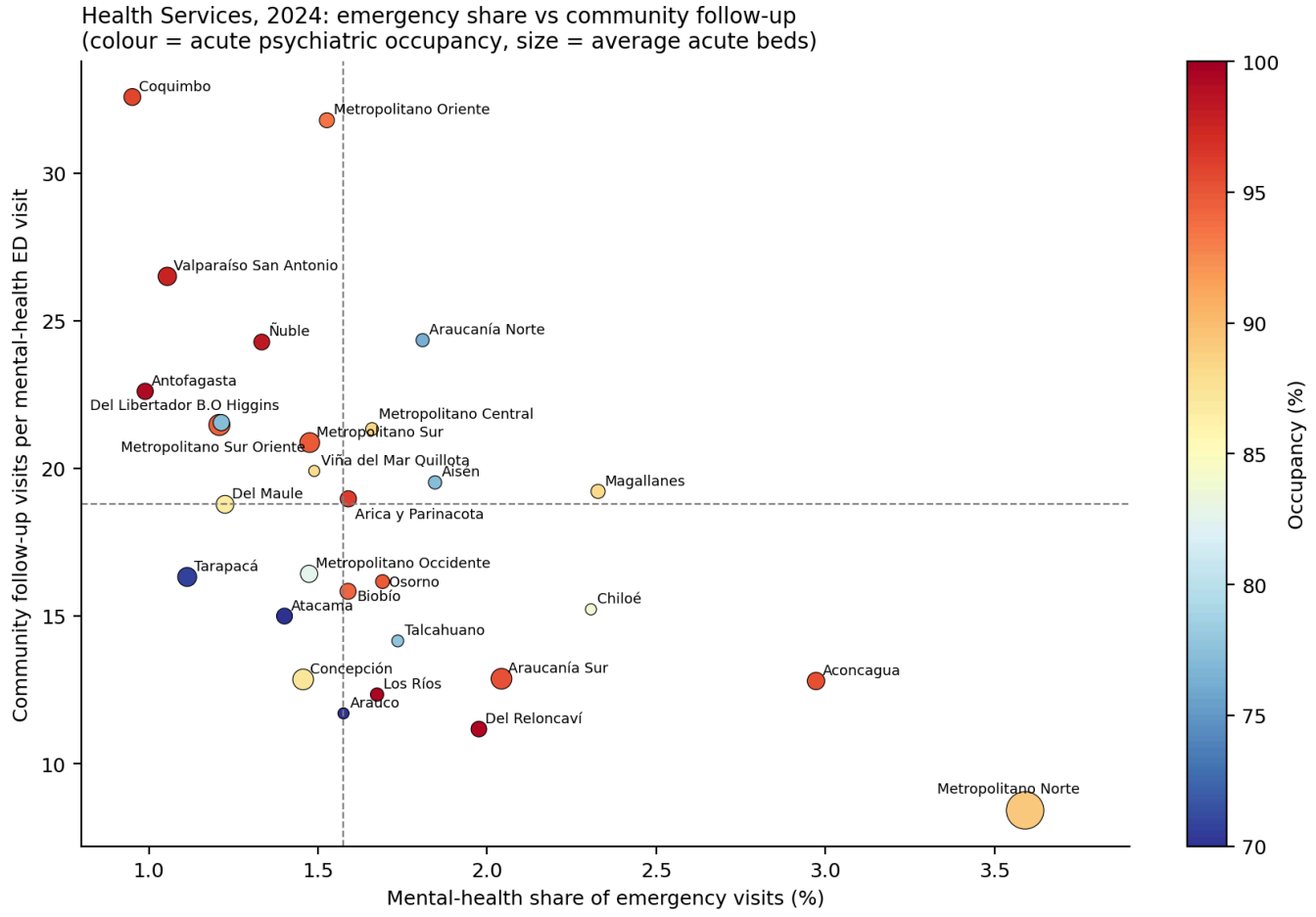
Concentration in specialised EDs. Instituto Psiquiátrico Dr. José Horwitz Barak receives 54.6% of Metropolitano Norte's mental-health visits and Hospital Psiquiátrico Dr. Philippe Pinel 60.6% of Aconcagua's. Without these two EDs the Services' shares fall from 3.59% to 1.66% and from 2.97% to 1.19%. Their high values describe how care is organised, with a referral centre that receives patients from other areas; they do not show higher need among local residents. Metropolitano Norte also has 231 of the 1,130 acute psychiatric beds.

Values that point to recording differences. Araucanía Norte records 0.7 ED hospitalisations per 100 mental-health visits, against a median of 21.1. Arauco's acute occupancy is 41.5%, against a median of 89.2%. These gaps are large enough to check how the data were recorded before reading them as differences in care.

Relationships between indicators. Across the 29 Services, the mental-health share and follow-up per ED visit move in opposite directions (Spearman −0.56). Part of this is built into the measures, since mental-health ED visits appear in both. Occupancy is almost unrelated to the share (−0.02) and to follow-up (0.10). High occupancy is therefore not explained by more mental-health emergency activity.

5.3 Screen: where to look first

Each Service gets a flag when it is above the median share of mental-health ED visits (excluding the two psychiatric EDs), below the median follow-up per ED visit, or at or above 90% acute occupancy. The 90% cut-off is a reading aid, not an official standard. Median-based flags mark about half the Services by design.



Dashed lines: medians of the 29 Services. Descriptive screen; not a ranking of quality or need.

Figure 3. Emergency share against community follow-up; colour = acute occupancy, size = average acute beds.

Flags	Services	Suggested first review
All three	Biobío, Araucanía Sur, Los Ríos, Osorno, Del Reloncaví	Community follow-up first, since low follow-up coincides with a high emergency share and full beds; then bed management
Two	Arica y Parinacota (emergency, hospital); Aconcagua (community, hospital); Metropolitano Norte, Talcahuano, Arauco, Chiloé (emergency, community)	The two flagged areas; for Metropolitano Norte and Aconcagua, first the referral role of the psychiatric ED
None	Viña del Mar Quillota, Del Libertador B. O'Higgins, Del Maule	No priority from this screen

The occupancy flag depends on the cut-off: 20 Services at 85%, 14 at 90%, 9 at 95%. The five Services with all three flags are contiguous, from Biobío south to Del Reloncaví. This location pattern is an observation; the data do not explain it.

5.4 Reconciling W1 figures

The W1 ratio, (causes 35+37+39+40) / psychiatric bed-days in adult short stay, child/adolescent short stay and adult long stay, reproduces the W1 report: Del Maule 1.83, Talcahuano 1.62. With C1's acute bed definition Del Maule falls to 1.17 and fifth place, because W1 left out its psychiatric intensive-care beds. The W1 ratio was undefined for Tarapacá, Antofagasta, Viña del Mar Quillota and Araucanía Norte, which have acute psychiatric beds only in areas W1 excluded. The W1 claim that 18 of 29 Services had no child/adolescent psychiatric beds was reversed. In REM20 2024, 18 Services report child/adolescent short-stay beds. Seven report no acute child/adolescent area: Viña del Mar Quillota, Aconcagua, Ñuble, Los Ríos, Osorno, Arauco and Chiloé. That means "not reported in REM20", not "no service": children may be referred to another Service's hospital, and REM20 does not record referrals. The ratio is not used for conclusions. Visits and bed-days are different units, and most visits do not need a bed, so a value above 1 does not show a shortage.

6. Conclusions and limits

6.1 What the evidence supports

Observed. Services differ several-fold in community follow-up per mental-health emergency visit (8.4 to 32.6) and in acute occupancy (41.5% to 99.6%). Five Services combine a high emergency share, low follow-up and high occupancy. Two Services' high emergency shares come from psychiatric referral EDs.

Interpretation. For the five three-flag Services, the pattern is consistent with community follow-up capacity that is low relative to crisis activity. This makes community follow-up a reasonable first area to review, with bed management second.

Hypotheses, not tested here. Low follow-up could lead to more crises reaching the ED. High occupancy could come from long stays caused by limited discharge options.

Alternative explanations.

- Differences in recording practice: SADU cause coding, REM A06 completeness.
- Referral flows between Services, which move visits and admissions away from the patient's own Service.
- Private-sector care, which is not observed.
- Differences in population size and age structure, which counts per Service do not adjust for.

What the analysis cannot establish:

- need or risk in the population;
- service quality;
- waiting time for a bed;
- causal effects;
- whether a person seen in the ED later received follow-up.

6.2 Implications for the decision and evidence still needed

A Health Service can use the profile and screen to pick an area to review first. Before acting on it, it needs:

1. records of the decision to hospitalise and the admission for each ED patient, like the Horwitz study, to measure waiting;
2. the population registered with FONASA or in primary care per Service, to compute rates;
3. REM20 and SADU for 2021–2023 (already available) to test whether the positions persist;
4. confirmation from DEIS or the Service of the recording anomalies in Araucanía Norte and Arauco.

7. Continuity: machine learning and dashboard

7.1 Proposed machine-learning task

Task. Cluster Health Services into profiles using the indicators in §4.5. Following the feedback, this starts with profiles and considers clustering to compare similar networks.

- **Unit of analysis:** Health Service × year, 2021–2024 (29 × 4 = 116 rows). SADU 2021–2023, REM20 since 2014 and REM Serie A 2022–2026 are already in the course data. 29 rows for one year are too few for a stable clustering.
- **Inputs:** the five indicators, plus the share of follow-up in primary care and the child/adolescent share of acute beds.
- **Output:** groups of Services with similar profiles. There are no outcome labels. Clusters are described, not validated against a label.
- **Decision value:** a Service compares itself with peers of similar profile rather than with the national median.
- **To review next stage:** whether the 2021–2023 cause codes and REM A06 sections match 2024 (Consistent); whether the psychiatric EDs need separate treatment; and scaling with so few units.

A supervised task needs an outcome the data define. One candidate is a hospital-area month with acute occupancy ≥ 95%, predicted from the previous months' ED mental-health visits and follow-up volume, which are known before the month starts. That is a prospective setting; same-month SADU counts would leak information. It is listed for instructor review, not committed. The W1 stress ratio does not define an outcome and is dropped.

7.2 Proposed dashboard

- **Users:** Health Service mental-health unit; MINSAL network division.
- **Decision supported:** which area to review first, and whether a Service's position persists over time.
- **Core indicators:** the five indicators of §4.5, each shown with the median of all Services and of the Service's cluster.
- **Views:** (1) screen scatter as in Figure 3; (2) Service profile page with monthly visits, occupancy and follow-up; (3) data-quality panel showing days reported, zero-bed months and codes changed.
- **Filters:** year, month, SADU cause, age band, facility type (hospital ED versus SAPU/SAR/SUR), with or without psychiatric EDs.
- **From the EDA:** the psychiatric-ED toggle comes from §5.2; the quality panel comes from the Araucanía Norte and Arauco anomalies; the waiting indicator is left out because SADU does not record it.

8. Process record

8.1 Decisions and checks

Date / stage	Decision	Alternative considered	Reason and evidence	Check performed	Result / location
23-Sep, W1	Aggregate SADU and REM20 to Health Service	Facility-level join	REM20 has no primary-care EDs; SADU has no beds	Facility match test	Kept in C1; notebook §4–5

Date / stage	Decision	Alternative considered	Reason and evidence	Check performed	Result / location
29-Sep, C1	Mental-health visits = SADU 36; self-harm (35) separate	W1 set 35+37+39+40	36 = 37..41 on all facility-days	Hierarchy check	Notebook §2.2
29-Sep, C1	Use SADU 42 as ED hospitalisations for mental disorders	Ignore; treat 42 as a visit cause	25 = 7+8+22+23+32+33+42 on all facility-days	Hierarchy check on raw 2024 file and extract	Notebook §2.2
29-Sep, C1	Waiting not measured	Use causes 27/28	All zero in 2024	Totals by cause	Notebook §2.2
29-Sep, C1	Acute psychiatric areas only	W1 areas (incl. long stay)	Mean stay 17–26 days vs 34–1,379 in excluded areas	Area profile	Notebook §6.1; report §5.4
29-Sep, C1	Lookup on both code systems	Legacy code only (W1 notebook)	31 facilities report modern codes	N:1 cardinality, unmatched = 0	Notebook §4
29-Sep, C1	Keep SADU 2024 Health Service assignment	Use 2026 registry	Registry is later than the visits	2 facilities differ, 0.12% of visits	Notebook §4
29-Sep, C1	Emergency flag excludes psychiatric EDs	Raw share	Two EDs hold >50% of their Service’s visits	Top-facility share	Notebook §9.2
29-Sep, C1	Remove ENS “15%” target and “2.4%” budget claims	Keep W1 text	Not found in the cited documents	Read ENS 2022 p. 35–37 and U. de Chile report	Report §2.2

8.2 Instructor feedback and response

Advice (W1 feedback, received before C1)	Response	Change and evidence
Use policy as context, not as proof of legal noncompliance	Accepted	Law 21.331 and ENS 2030 used as context; unsupported target and budget figures removed (§2.2)
Take the reader from the wider problem to the case; define population, territory, period; visits are not people; one year is not a seasonal pattern	Accepted	§2.1, §5.1 start from all ED visits; “visits, not people” stated throughout; no seasonal claim
Explain what the comparison helps decide: compare community care, emergency visits and hospital use	Accepted	REM A06 follow-up added; decision and screen in §3 and §5.3
Resolve 640 vs 609 facilities and the differing territorial labels	Accepted	§4.3: 609 legacy + 31 modern codes; labels mapped by code
Check network membership with the DEIS historical lists	Adapted	Download blocked from our environment; 2026 registry used and the gap stated (§4.3)
Record each person’s work and learning; tidy instructions	Accepted	§8.3–8.5; README rewritten
Show visits, bed-days and occupancy separately; ratio > 1 does not show a shortage	Accepted	Separate indicators; ratio kept only for reconciliation (§5.4)
Zero pediatric bed-days: check reporting and referral first; keep the actual age label	Accepted	“Not reported in REM20” wording; 15–64 label kept (§5.1, §5.4)
Start with profiles, cluster later if useful; classification needs a defined outcome	Accepted	§7.1

Advice (W1 feedback, received before C1)	Response	Change and evidence
REM A06 as a possible addition; try a small comparison	Accepted	REM A06 section A.1 included as the community indicator

8.3 Individual contributions

Member	Concrete contribution	Evidence location
Benjamín Buzeta	W1: documentary context and comparison of the two lines; W1 report. C1: ran the C1 revision with Claude against the feedback; supplied the DEIS downloads; checked the policy claims (ENS 2030, Law 21.331, financing report, Horwitz study) and removed the unsupported ones; wrote the problem, seven-step map, conclusions and Process record	Report §2, §3, §6, §8; git commit of 24-Sep
Oscar Pichuman Millape	W1: extraction of reduced datasets, W1 notebook (executed on his machine), join tests and repository structure. C1: reviewed the extraction filters and the profiling results (cause hierarchy, 609 + 31 facility codes, territorial labels); checked the acute-bed definition against the REM20 area profile	`analysis/workshop1.ipynb`; `analysis/extract_c1_inputs.py`; notebook §2–6; git commits of 24-Sep
Felipe Alfredo Tajan Gebauer	W1: reviewed the comparison matrix and drafted the questions for instructor feedback. C1: selected the REM A06 section A.1 codes from the SA_24 dictionary for the community follow-up indicator; reviewed the Health Service profiles, the screen and its cut-off sensitivity; wrote the machine-learning and dashboard proposal; prepared the slides and presentation script	Report §4.5, §5.2–5.3, §7; notebook §7, §9–10; `presentation/`

8.4 AI use and verification

Tool and purpose	What was accepted, changed or rejected	How it was checked
W1: Antigravity CLI with Gemini 3.8 Flash; code scaffolding, translation, plot styling	Rejected a direct SADU–REM20 row join. The “15%” target and “2.4%” budget statements in the W1 report were not supported and have been removed in C1	Units of observation; the source documents themselves
C1: Claude (Anthropic); extraction script, notebook code, report and slides drafting	Accepted after checks: cause hierarchy, code lookup, indicator code. Changed: the emergency flag, after the concentration check showed two psychiatric EDs driving it. Prompt excerpt: “update the W1 project for C1 based on the feedback [...] keep changes minimal”	Every figure in this report is produced by the executed notebook; totals by Service reconcile with national totals; policy claims checked against the documents in §9

8.5 Individual reflection

Member	Contribution	Learning gain	Unresolved uncertainty
Benjamín Buzeta	Reframed the question around the decision a Health Service faces and removed claims the sources do not support	A policy objective, a numeric target and an indicator the data can measure are three different things; only the third can be tested with DEIS data	Whether the high follow-up and occupancy contrasts reflect care or recording practice, since REM A06 completeness per Service was not audited
Oscar Pichuman Millape	Built the W1 extraction and joins that C1 reuses, and checked the C1 profiling results	Summing SADU rows without checking the cause hierarchy double-counts visits; the hierarchy check changed which causes the analysis uses	How much of each Service's emergency activity comes from patients living in other Services, which none of the sources records
Felipe Alfredo Tajan Gebauer	Added community follow-up as the third part of the comparison and wrote the continuity proposal	A ratio between two sources describes relative volume, not coverage: follow-up visits per emergency visit cannot say whether the same people were followed up	Whether 29 Services × 4 years gives stable clusters, and whether REM A06 section A.1 captures all COSAM activity in every Service

9. References

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Annex

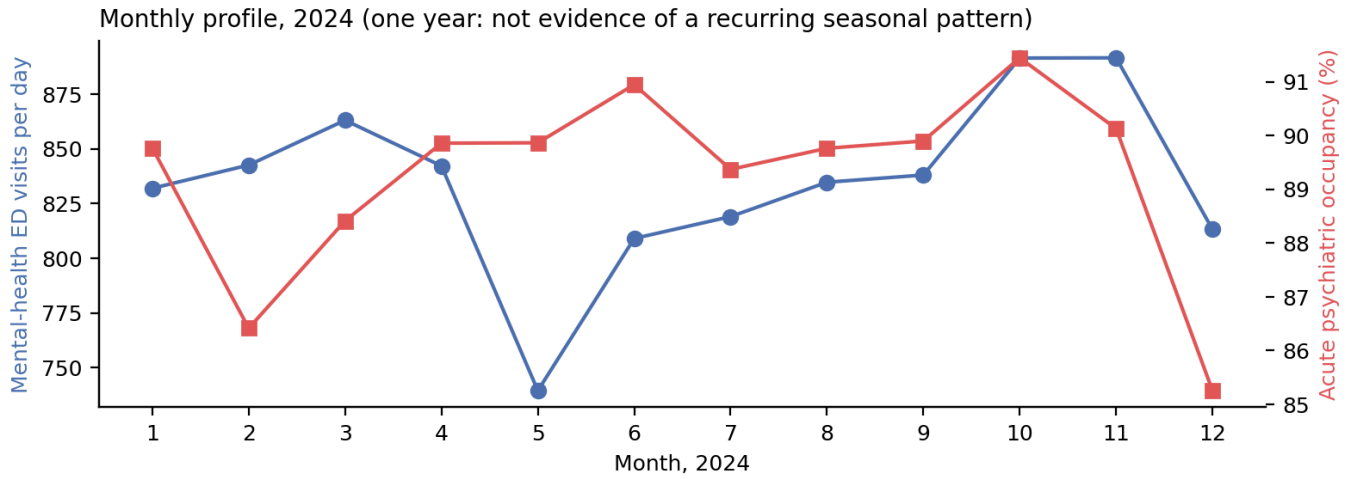


Figure 4. Monthly mental-health ED visits per day and acute psychiatric occupancy, 2024.

Table A1. Health Service profile, 2024 (from `data/health_service_profile_2024.csv`; E/C/H = emergency, community and hospital flags).

Health Service	MH ED visits	MH share %	excl. psych ED %	ED hosp. /100	Acute beds	Occupancy %	Follow-up / ED visit	E	C	H
Aconcagua	7,456	2.97	1.19	9.0	40.0	95.4	12.8		✓	✓
Aisén	2,616	1.85	1.85	31.5	18.1	77.2	19.5	✓		
Antofagasta	5,252	0.99	0.99	25.1	33.4	99.2	22.6			✓
Araucanía Norte	3,922	1.81	1.81	0.7	17.6	76.2	24.3	✓		
Araucanía Sur	19,498	2.04	2.04	11.0	62.0	95.3	12.9	✓	✓	✓
Arauco	3,966	1.58	1.58	17.4	9.0	41.5	11.7	✓	✓	
Arica y Parinacota	3,652	1.59	1.59	24.0	33.0	96.1	19.0	✓		✓
Atacama	5,187	1.40	1.40	22.0	31.1	68.6	15.0		✓	
Biobío	9,213	1.59	1.59	9.5	32.0	94.2	15.8	✓	✓	✓
Chiloé	4,581	2.31	2.31	22.1	9.0	83.9	15.2	✓	✓	
Concepción	13,875	1.46	1.46	20.7	61.9	87.2	12.9		✓	
Coquimbo	7,004	0.95	0.95	21.1	36.9	95.9	32.6			✓
Del Libertador B. O'Higgins	15,161	1.21	1.21	12.4	33.1	77.2	21.6			
Del Maule	20,661	1.23	1.23	24.8	42.3	86.9	18.8			
Del Reloncaví	10,877	1.98	1.98	19.7	30.0	99.4	11.2	✓	✓	✓
Los Ríos	9,569	1.67	1.67	22.1	18.3	99.6	12.3	✓	✓	✓
Magallanes	4,976	2.33	2.33	21.9	22.2	88.2	19.2	✓		
Metropolitano Central	9,985	1.66	1.66	19.2	15.8	88.4	21.3	✓		
Metropolitano Norte	32,983	3.59	1.66	7.8	230.7	89.2	8.4	✓	✓	
Metropolitano Occidente	22,129	1.47	1.47	6.0	38.6	82.6	16.4		✓	

Health Service	MH ED visits	MH share %	excl. psych ED %	ED hosp. /100	Acute beds	Occupancy %	Follow-up / ED visit	E	C	H
Metropolitano Oriente	9,019	1.53	1.53	43.9	26.7	93.5	31.8			✓
Metropolitano Sur	17,385	1.48	1.48	20.0	55.0	94.8	20.9			✓
Metropolitano Sur Oriente	16,514	1.21	1.21	31.1	64.8	94.4	21.5			✓
Osorno	5,367	1.69	1.69	17.0	21.0	94.8	16.2	✓	✓	✓
Talcahuano	9,155	1.74	1.74	14.0	12.0	77.8	14.2	✓	✓	
Tarapacá	5,610	1.11	1.11	35.5	49.7	70.7	16.3		✓	
Valparaíso San Antonio	6,443	1.05	1.05	31.6	45.9	97.7	26.5			✓
Viña del Mar Quillota	15,125	1.49	1.49	21.6	8.7	88.0	19.9			
Ñuble	8,250	1.33	1.33	27.9	31.0	98.2	24.3			✓